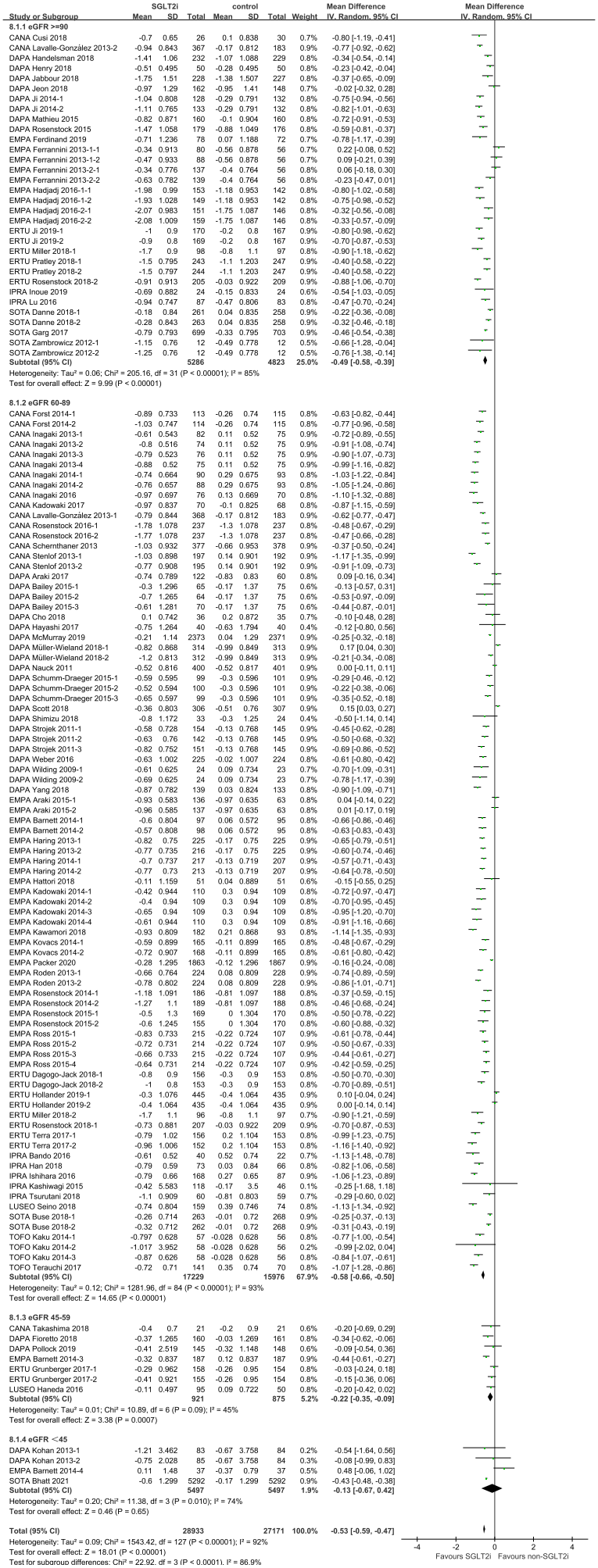

SGLT2i
control
Mean Difference
Mean Difference
Study or Subgroup Mean SD Total Mean SD Total Weight IV, Random, 95% CI
IV, Random, 95% CI
8.1.1 eGFR >=90
CANA Cusi 2018 -0.7 0.65 26 0.1 0.838 30 0.7% -0.80 [-1.19, -0.41]
CANA Lavalle-González 2013-2 -0.94 0.843 367 -0.17 0.812 183 0.9% -0.77 [-0.92, -0.62]
DAPA Handelsman 2018 -1.41 1.06 232 -1.07 1.088 229 0.8% -0.34 [-0.54, -0.14]
DAPA Henry 2018 -0.51 0.495 50 -0.28 0.495 50 0.8% -0.23 [-0.42, -0.04]
DAPA Jabbour 2018 -1.75 1.51 228 -1.38 1.507 227 0.8% -0.37 [-0.65, -0.09]
DAPA Jeon 2018 -0.97 1.29 162 -0.95 1.41 148 0.7% -0.02 [-0.32, 0.28]
DAPA Ji 2014-1 -1.04 0.808 128 -0.29 0.791 132 0.8% -0.75 [-0.94, -0.56]
DAPA Ji 2014-2 -1.11 0.765 133 -0.29 0.791 132 0.8% -0.82 [-1.01, -0.63]
DAPA Mathieu 2015 -0.82 0.871 160 -0.1 0.904 160 0.8% -0.72 [-0.91, -0.53]
DAPA Rosenstock 2015 -1.47 1.058 179 -0.88 1.049 176 0.8% -0.59 [-0.81, -0.37]
EMPA Ferdinand 2019 -0.71 1.236 78 0.07 1.188 72 0.7% -0.78 [-1.17, -0.39]
EMPA Ferrannini 2013-1-1 -0.34 0.913 80 -0.56 0.878 56 0.7% 0.22 [-0.08, 0.52]
EMPA Ferrannini 2013-1-2 -0.47 0.933 88 -0.56 0.878 56 0.7% 0.09 [-0.21, 0.39]
EMPA Ferrannini 2013-2-1 -0.34 0.776 137 -0.4 0.764 56 0.8% 0.06 [-0.18, 0.30]
EMPA Ferrannini 2013-2-2 -0.63 0.782 139 -0.4 0.764 56 0.8% -0.23 [-0.47, 0.01]
EMPA Hadjadj 2016-1-1 -1.98 0.99 153 -1.18 0.953 142 0.8% -0.80 [-1.02, -0.58]
EMPA Hadjadj 2016-1-2 -1.93 1.028 149 -1.18 0.953 142 0.8% -0.75 [-0.98, -0.52]
EMPA Hadjadj 2016-2-1 -2.07 0.983 151 -1.75 1.087 146 0.8% -0.32 [-0.56, -0.08]
EMPA Hadjadj 2016-2-2 -2.08 1.009 159 -1.75 1.087 146 0.8% -0.33 [-0.57, -0.09]
ERTU Ji 2019-1 -1 0.9 170 -0.2 0.8 167 0.9% -0.80 [-0.98, -0.62]
ERTU Ji 2019-2 -0.9 0.8 169 -0.2 0.8 167 0.9% -0.70 [-0.87, -0.53]
ERTU Miller 2018-1 -1.7 0.9 98 -0.8 1.1 97 0.8% -0.90 [-1.18, -0.62]
ERTU Pratley 2018-1 -1.5 0.795 243 -1.1 1.203 247 0.9% -0.40 [-0.58, -0.22]
ERTU Pratley 2018-2 -1.5 0.797 244 -1.1 1.203 247 0.9% -0.40 [-0.58, -0.22]
ERTU Rosenstock 2018-2 -0.91 0.913 205 -0.03 0.922 209 0.9% -0.88 [-1.06, -0.70]
IPRA Inoue 2019 -0.69 0.882 24 -0.15 0.833 24 0.6% -0.54 [-1.03, -0.05]
IPRA Lu 2016 -0.94 0.747 87 -0.47 0.806 83 0.8% -0.47 [-0.70, -0.24]
SOTA Danne 2018-1 -0.18 0.84 261 0.04 0.835 258 0.9% -0.22 [-0.36, -0.08]
SOTA Danne 2018-2 -0.28 0.843 263 0.04 0.835 258 0.9% -0.32 [-0.46, -0.18]
SOTA Garg 2017 -0.79 0.793 699 -0.33 0.795 703 0.9% -0.46 [-0.54, -0.38]
SOTA Zambrowicz 2012-1 -1.15 0.76 12 -0.49 0.778 12 0.5% -0.66 [-1.28, -0.04]
SOTA Zambrowicz 2012-2 -1.25 0.76 12 -0.49 0.778 12 0.5% -0.76 [-1.38, -0.14]
Subtotal (95% CI) 5286 4823 25.0% -0.49 [-0.58, -0.39]
Heterogeneity: Tau² = 0.06; Chi² = 205.16, df = 31 (P < 0.00001); I² = 85%
Test for overall effect: Z = 9.99 (P < 0.00001)
8.1.2 eGFR 60-89
CANA Forst 2014-1 -0.89 0.733 113 -0.26 0.74 115 0.8% -0.63 [-0.82, -0.44]
CANA Forst 2014-2 -1.03 0.747 114 -0.26 0.74 115 0.8% -0.77 [-0.96, -0.58]
CANA Inagaki 2013-1 -0.61 0.543 82 0.11 0.52 75 0.9% -0.72 [-0.89, -0.55]
CANA Inagaki 2013-2 -0.8 0.516 74 0.11 0.52 75 0.9% -0.91 [-1.08, -0.74]
CANA Inagaki 2013-3 -0.79 0.523 76 0.11 0.52 75 0.9% -0.90 [-1.07, -0.73]
CANA Inagaki 2013-4 -0.88 0.52 75 0.11 0.52 75 0.9% -0.99 [-1.16, -0.82]
CANA Inagaki 2014-1 -0.74 0.664 90 0.29 0.675 93 0.8% -1.03 [-1.22, -0.84]
CANA Inagaki 2014-2 -0.76 0.657 88 0.29 0.675 93 0.8% -1.05 [-1.24, -0.86]
CANA Inagaki 2016 -0.97 0.697 76 0.13 0.669 70 0.8% -1.10 [-1.32, -0.88]
CANA Kadowaki 2017 -0.97 0.837 70 -0.1 0.825 68 0.8% -0.87 [-1.15, -0.59]
CANA Lavalle-González 2013-1 -0.79 0.844 368 -0.17 0.812 183 0.9% -0.62 [-0.77, -0.47]
CANA Rosenstock 2016-1 -1.78 1.078 237 -1.3 1.078 237 0.8% -0.48 [-0.67, -0.29]
CANA Rosenstock 2016-2 -1.77 1.078 237 -1.3 1.078 237 0.8% -0.47 [-0.66, -0.28]
CANA Schernthaner 2013 -1.03 0.932 377 -0.66 0.953 378 0.9% -0.37 [-0.50, -0.24]
CANA Stenlof 2013-1 -1.03 0.898 197 0.14 0.901 192 0.9% -1.17 [-1.35, -0.99]
CANA Stenlof 2013-2 -0.77 0.908 195 0.14 0.901 192 0.9% -0.91 [-1.09, -0.73]
DAPA Araki 2017 -0.74 0.789 122 -0.83 0.83 60 0.8% 0.09 [-0.16, 0.34]
DAPA Bailey 2015-1 -0.3 1.296 65 -0.17 1.37 75 0.6% -0.13 [-0.57, 0.31]
DAPA Bailey 2015-2 -0.7 1.265 64 -0.17 1.37 75 0.6% -0.53 [-0.97, -0.09]
DAPA Bailey 2015-3 -0.61 1.281 70 -0.17 1.37 75 0.6% -0.44 [-0.87, -0.01]
DAPA Cho 2018 0.1 0.742 36 0.2 0.872 35 0.7% -0.10 [-0.48, 0.28]
DAPA Hayashi 2017 -0.75 1.264 40 -0.63 1.794 40 0.4% -0.12 [-0.80, 0.56]
DAPA McMurray 2019 -0.21 1.14 2373 0.04 1.29 2371 0.9% -0.25 [-0.32, -0.18]
DAPA Müller-Wieland 2018-1 -0.82 0.868 314 -0.99 0.849 313 0.9% 0.17 [0.04, 0.30]
DAPA Müller-Wieland 2018-2 -1.2 0.813 312 -0.99 0.849 313 0.9% -0.21 [-0.34, -0.08]
DAPA Nauck 2011 -0.52 0.813 400 -0.52 0.817 401 0.9% 0.00 [-0.11, 0.11]
DAPA Schumm-Draeger 2015-1 -0.59 0.595 99 -0.3 0.596 101 0.9% -0.29 [-0.46, -0.12]
DAPA Schumm-Draeger 2015-2 -0.52 0.594 100 -0.3 0.596 101 0.9% -0.22 [-0.38, -0.06]
DAPA Schumm-Draeger 2015-3 -0.65 0.597 99 -0.3 0.596 101 0.9% -0.35 [-0.52, -0.18]
DAPA Scott 2018 -0.36 0.803 306 -0.51 0.76 307 0.9% 0.15 [0.03, 0.27]
DAPA Shimizu 2018 -0.8 1.172 33 -0.3 1.25 24 0.4% -0.50 [-1.14, 0.14]
DAPA Strojek 2011-1 -0.58 0.728 154 -0.13 0.768 145 0.9% -0.45 [-0.62, -0.28]
DAPA Strojek 2011-2 -0.63 0.76 142 -0.13 0.768 145 0.9% -0.50 [-0.68, -0.32]
DAPA Strojek 2011-3 -0.82 0.752 151 -0.13 0.768 145 0.9% -0.69 [-0.86, -0.52]
DAPA Weber 2016 -0.63 1.002 225 -0.02 1.007 224 0.8% -0.61 [-0.80, -0.42]
DAPA Wilding 2009-1 -0.61 0.625 24 0.09 0.734 23 0.7% -0.70 [-1.09, -0.31]
DAPA Wilding 2009-2 -0.69 0.625 24 0.09 0.734 23 0.7% -0.78 [-1.17, -0.39]
DAPA Yang 2018 -0.87 0.782 139 0.03 0.824 133 0.8% -0.90 [-1.09, -0.71]
EMPA Araki 2015-1 -0.93 0.583 136 -0.97 0.635 63 0.8% 0.04 [-0.14, 0.22]
EMPA Araki 2015-2 -0.96 0.585 137 -0.97 0.635 63 0.8% 0.01 [-0.17, 0.19]
EMPA Barnett 2014-1 -0.6 0.804 97 0.06 0.572 95 0.8% -0.66 [-0.86, -0.46]
EMPA Barnett 2014-2 -0.57 0.808 98 0.06 0.572 95 0.8% -0.63 [-0.83, -0.43]
EMPA Haring 2013-1 -0.82 0.75 225 -0.17 0.75 225 0.9% -0.65 [-0.79, -0.51]
EMPA Haring 2013-2 -0.77 0.735 216 -0.17 0.75 225 0.9% -0.60 [-0.74, -0.46]
EMPA Haring 2014-1 -0.7 0.737 217 -0.13 0.719 207 0.9% -0.57 [-0.71, -0.43]
EMPA Haring 2014-2 -0.77 0.73 213 -0.13 0.719 207 0.9% -0.64 [-0.78, -0.50]
EMPA Hattori 2018 -0.11 1.159 51 0.04 0.889 51 0.6% -0.15 [-0.55, 0.25]
EMPA Kadowaki 2014-1 -0.42 0.944 110 0.3 0.94 109 0.8% -0.72 [-0.97, -0.47]
EMPA Kadowaki 2014-2 -0.4 0.94 109 0.3 0.94 109 0.8% -0.70 [-0.95, -0.45]
EMPA Kadowaki 2014-3 -0.65 0.94 109 0.3 0.94 109 0.8% -0.95 [-1.20, -0.70]
EMPA Kadowaki 2014-4 -0.61 0.944 110 0.3 0.94 109 0.8% -0.91 [-1.16, -0.66]
EMPA Kawamori 2018 -0.93 0.809 182 0.21 0.868 93 0.8% -1.14 [-1.35, -0.93]
EMPA Kovacs 2014-1 -0.59 0.899 165 -0.11 0.899 165 0.8% -0.48 [-0.67, -0.29]
EMPA Kovacs 2014-2 -0.72 0.907 168 -0.11 0.899 165 0.8% -0.61 [-0.80, -0.42]
EMPA Packer 2020 -0.28 1.295 1863 -0.12 1.296 1867 0.9% -0.16 [-0.24, -0.08]
EMPA Roden 2013-1 -0.66 0.764 224 0.08 0.809 228 0.9% -0.74 [-0.89, -0.59]
EMPA Roden 2013-2 -0.78 0.802 224 0.08 0.809 228 0.9% -0.86 [-1.01, -0.71]
EMPA Rosenstock 2014-1 -1.18 1.091 186 -0.81 1.097 188 0.8% -0.37 [-0.59, -0.15]
EMPA Rosenstock 2014-2 -1.27 1.1 189 -0.81 1.097 188 0.8% -0.46 [-0.68, -0.24]
EMPA Rosenstock 2015-1 -0.5 1.3 169 0 1.304 170 0.8% -0.50 [-0.78, -0.22]
EMPA Rosenstock 2015-2 -0.6 1.245 155 0 1.304 170 0.8% -0.60 [-0.88, -0.32]
EMPA Ross 2015-1 -0.83 0.733 215 -0.22 0.724 107 0.9% -0.61 [-0.78, -0.44]
EMPA Ross 2015-2 -0.72 0.731 214 -0.22 0.724 107 0.9% -0.50 [-0.67, -0.33]
EMPA Ross 2015-3 -0.66 0.733 215 -0.22 0.724 107 0.9% -0.44 [-0.61, -0.27]
EMPA Ross 2015-4 -0.64 0.731 214 -0.22 0.724 107 0.9% -0.42 [-0.59, -0.25]
ERTU Dagogo-Jack 2018-1 -0.8 0.9 156 -0.3 0.9 153 0.8% -0.50 [-0.70, -0.30]
ERTU Dagogo-Jack 2018-2 -1 0.8 153 -0.3 0.9 153 0.8% -0.70 [-0.89, -0.51]
ERTU Hollander 2019-1 -0.3 1.076 445 -0.4 1.064 435 0.9% 0.10 [-0.04, 0.24]
ERTU Hollander 2019-2 -0.4 1.064 435 -0.4 1.064 435 0.9% 0.00 [-0.14, 0.14]
ERTU Miller 2018-2 -1.7 1.1 96 -0.8 1.1 97 0.7% -0.90 [-1.21, -0.59]
ERTU Rosenstock 2018-1 -0.73 0.881 207 -0.03 0.922 209 0.9% -0.70 [-0.87, -0.53]
ERTU Terra 2017-1 -0.79 1.02 156 0.2 1.104 153 0.8% -0.99 [-1.23, -0.75]
ERTU Terra 2017-2 -0.96 1.006 152 0.2 1.104 153 0.8% -1.16 [-1.40, -0.92]
IPRA Bando 2016 -0.61 0.52 40 0.52 0.74 22 0.7% -1.13 [-1.48, -0.78]
IPRA Han 2018 -0.79 0.59 73 0.03 0.84 66 0.8% -0.82 [-1.06, -0.58]
IPRA Ishihara 2016 -0.79 0.66 168 0.27 0.65 87 0.9% -1.06 [-1.23, -0.89]
IPRA Kashiwagi 2015 -0.42 5.583 118 -0.17 3.5 46 0.1% -0.25 [-1.68, 1.18]
IPRA Tsurutani 2018 -1.1 0.909 60 -0.81 0.803 59 0.7% -0.29 [-0.60, 0.02]
LUSEO Seino 2018 -0.74 0.804 159 0.39 0.746 74 0.8% -1.13 [-1.34, -0.92]
SOTA Buse 2018-1 -0.26 0.714 263 -0.01 0.72 268 0.9% -0.25 [-0.37, -0.13]
SOTA Buse 2018-2 -0.32 0.712 262 -0.01 0.72 268 0.9% -0.31 [-0.43, -0.19]
TOFO Kaku 2014-1 -0.797 0.628 57 -0.028 0.628 56 0.8% -0.77 [-1.00, -0.54]
TOFO Kaku 2014-2 -1.017 3.952 58 -0.028 0.628 56 0.2% -0.99 [-2.02, 0.04]
TOFO Kaku 2014-3 -0.87 0.626 58 -0.028 0.628 56 0.8% -0.84 [-1.07, -0.61]
TOFO Terauchi 2017 -0.72 0.71 141 0.35 0.74 70 0.8% -1.07 [-1.28, -0.86]
Subtotal (95% CI) 17229 15976 67.9% -0.58 [-0.66, -0.50]
Heterogeneity: Tau² = 0.12; Chi² = 1281.96, df = 84 (P < 0.00001); I² = 93%
Test for overall effect: Z = 14.65 (P < 0.00001)
8.1.3 eGFR 45-59
CANA Takashima 2018 -0.4 0.7 21 -0.2 0.9 21 0.6% -0.20 [-0.69, 0.29]
DAPA Fioretto 2018 -0.37 1.265 160 -0.03 1.269 161 0.8% -0.34 [-0.62, -0.06]
DAPA Pollock 2019 -0.41 2.519 145 -0.32 1.148 148 0.6% -0.09 [-0.54, 0.36]
EMPA Barnett 2014-3 -0.32 0.837 187 0.12 0.837 187 0.9% -0.44 [-0.61, -0.27]
ERTU Grunberger 2017-1 -0.29 0.962 158 -0.26 0.95 154 0.8% -0.03 [-0.24, 0.18]
ERTU Grunberger 2017-2 -0.41 0.921 155 -0.26 0.95 154 0.8% -0.15 [-0.36, 0.06]
LUSEO Haneda 2016 -0.11 0.497 95 0.09 0.722 50 0.8% -0.20 [-0.42, 0.02]
Subtotal (95% CI) 921 875 5.2% -0.22 [-0.35, -0.09]
Heterogeneity: Tau² = 0.01; Chi² = 10.89, df = 6 (P = 0.09); I² = 45%
Test for overall effect: Z = 3.38 (P = 0.0007)
8.1.4 eGFR <45
DAPA Kohan 2013-1 -1.21 3.462 83 -0.67 3.758 84 0.2% -0.54 [-1.64, 0.56]
DAPA Kohan 2013-2 -0.75 2.028 85 -0.67 3.758 84 0.3% -0.08 [-0.99, 0.83]
EMPA Barnett 2014-4 0.11 1.48 37 -0.37 0.79 37 0.5% 0.48 [-0.06, 1.02]
SOTA Bhatt 2021 -0.6 1.299 5292 -0.17 1.299 5292 0.9% -0.43 [-0.48, -0.38]
Subtotal (95% CI) 5497 5497 1.9% -0.13 [-0.67, 0.42]
Heterogeneity: Tau² = 0.20; Chi² = 11.38, df = 3 (P = 0.010); I² = 74%
Test for overall effect: Z = 0.46 (P = 0.65)
Total (95% CI) 28933 27171 100.0% -0.53 [-0.59, -0.47]
Heterogeneity: Tau² = 0.09; Chi² = 1543.42, df = 127 (P < 0.00001); I² = 92%
Test for overall effect: Z = 18.01 (P < 0.00001)
Test for subgroup differences: Chi² = 22.92, df = 3 (P < 0.0001), I² = 86.9%
-4 -2 0 2 4
Favours SGLT2i Favours non-SGLT2i